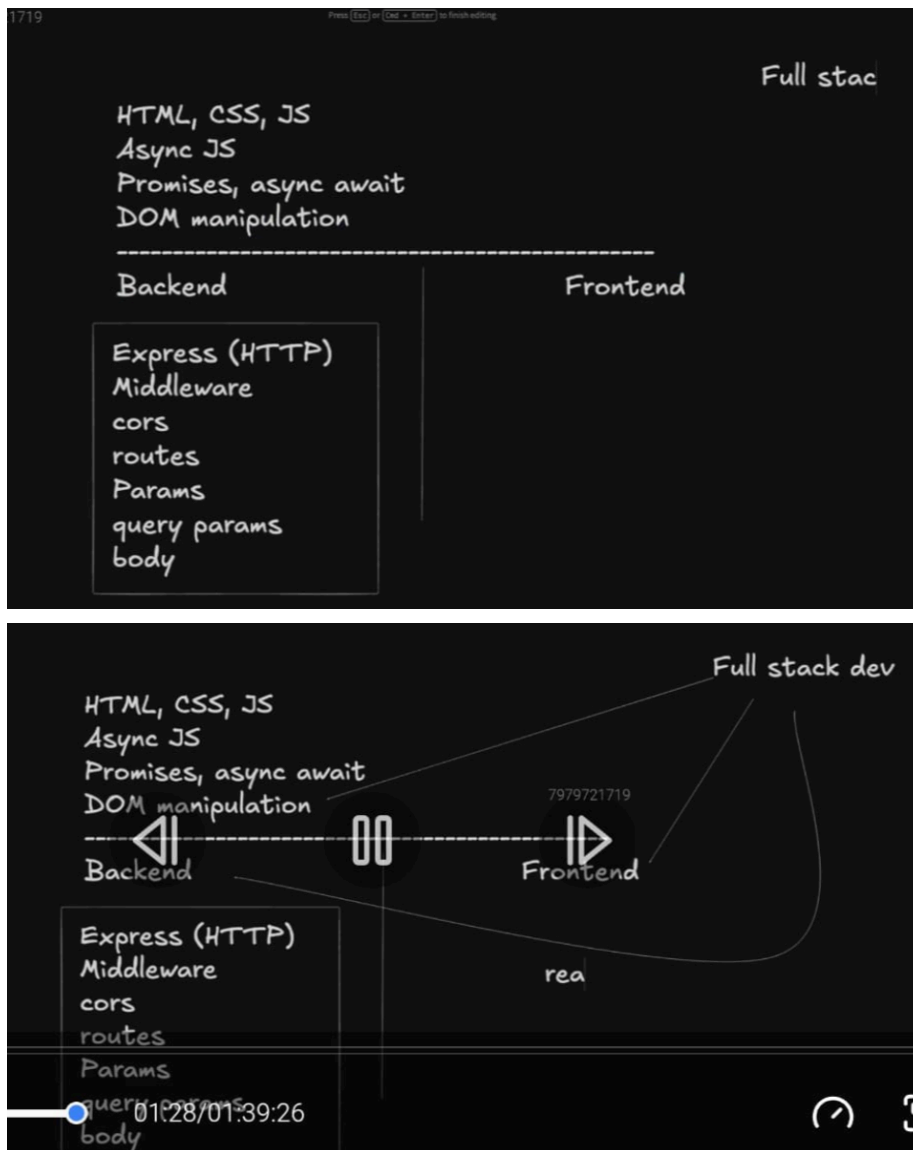


Notes: <https://projects.100xdevs.com/tracks/http-intro/Intro-to-HTTP-1>

1.

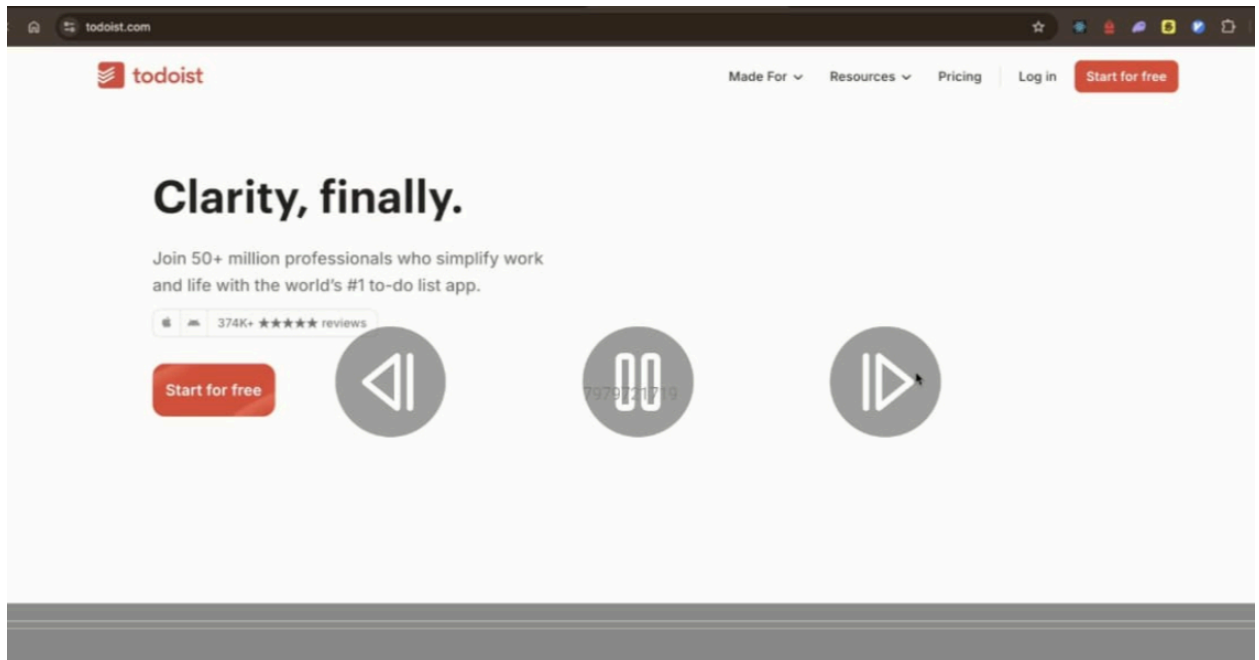
Harkirat explained what we learned till now in frontend and what we going to learn in backend



2. Today we will learn Intro to the HTTP Protocol, what does it solve

Exploring the network tab in the **chrome developer tools**

3. Later we will make backend of todoist application



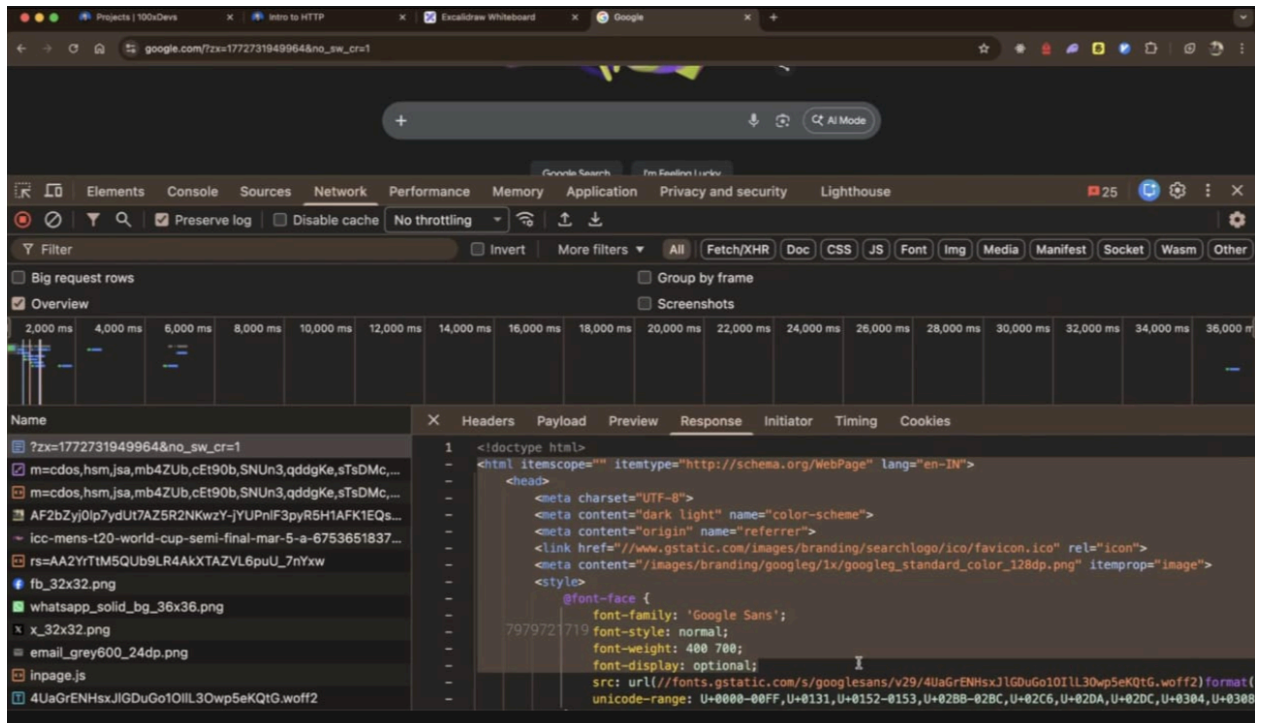
4. Around 2012–2013, cloud platforms like **AWS (Amazon Web Services)** made it much easier for small companies to manage data. Earlier, only big companies like **Amazon** could afford large servers and infrastructure. But with AWS, even startups like **Razorpay** can store their data in the cloud and deliver it to users efficiently at a much lower cost.



Data Centers



Now if we hit [google.com](https://www.google.com) then make request and response is coming from datacenter google server



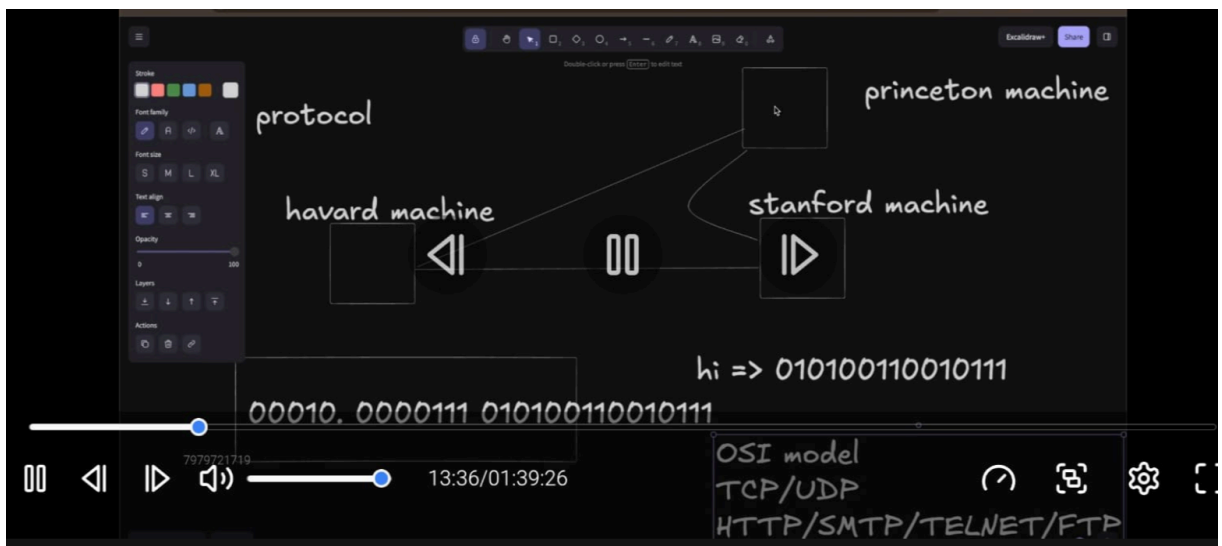
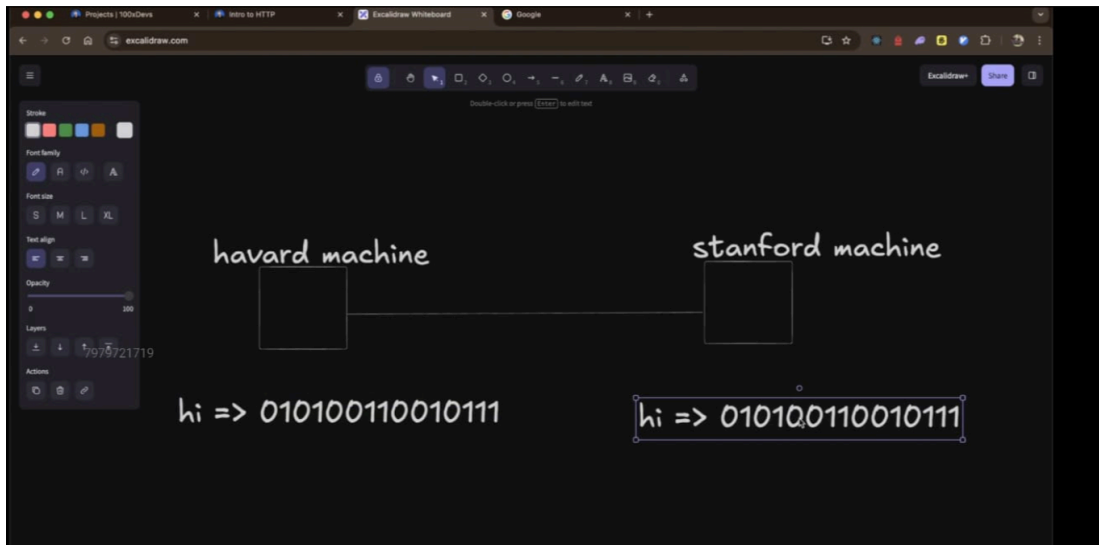
5. Why the HTTP Protocols?

When data is transferred from a machine at **Harvard** to a machine at **Stanford**, it travels in the form of **binary (0s and 1s)**.

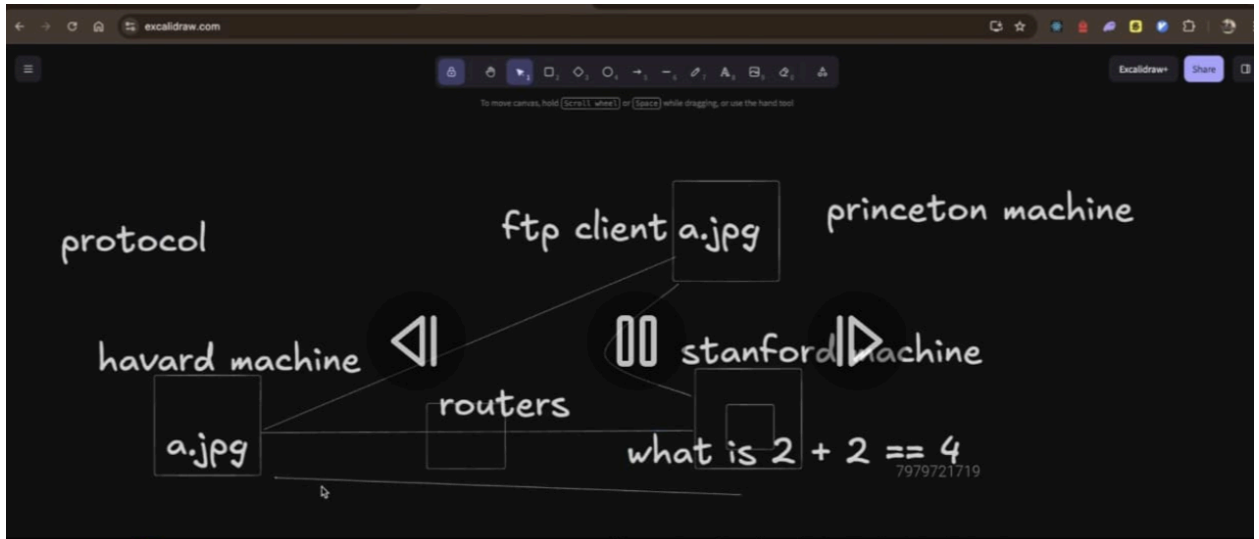
To ensure the data reaches the **correct destination and is interpreted correctly**, a set of rules is used. These rules are known as **network protocols**.

Protocols comes under OSI Model:

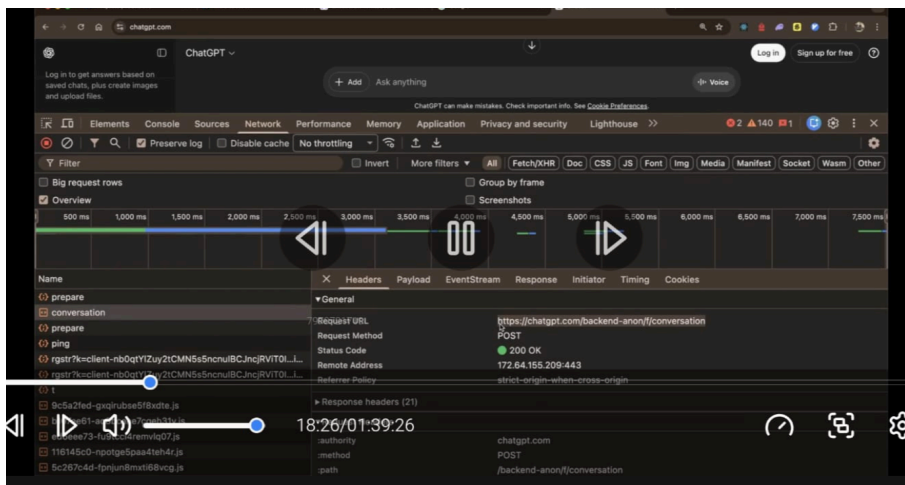
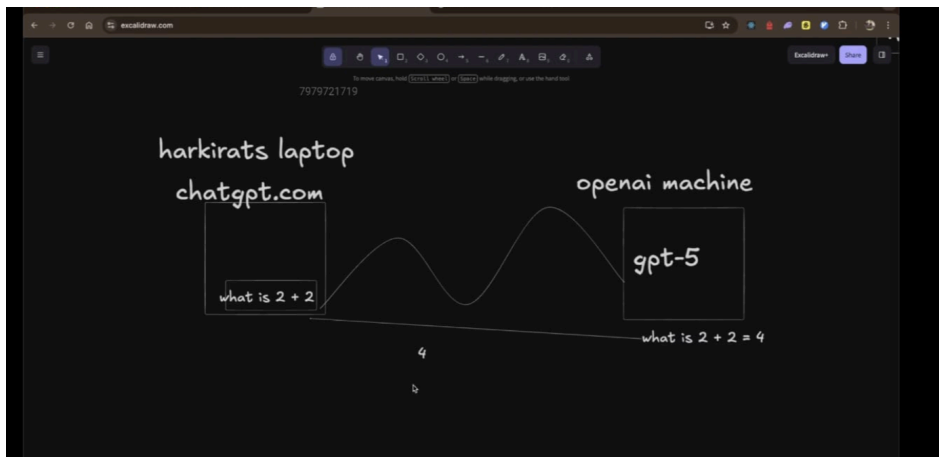
1. TCP/UDP
2. HTTP/SMTP/TELNET/FTP



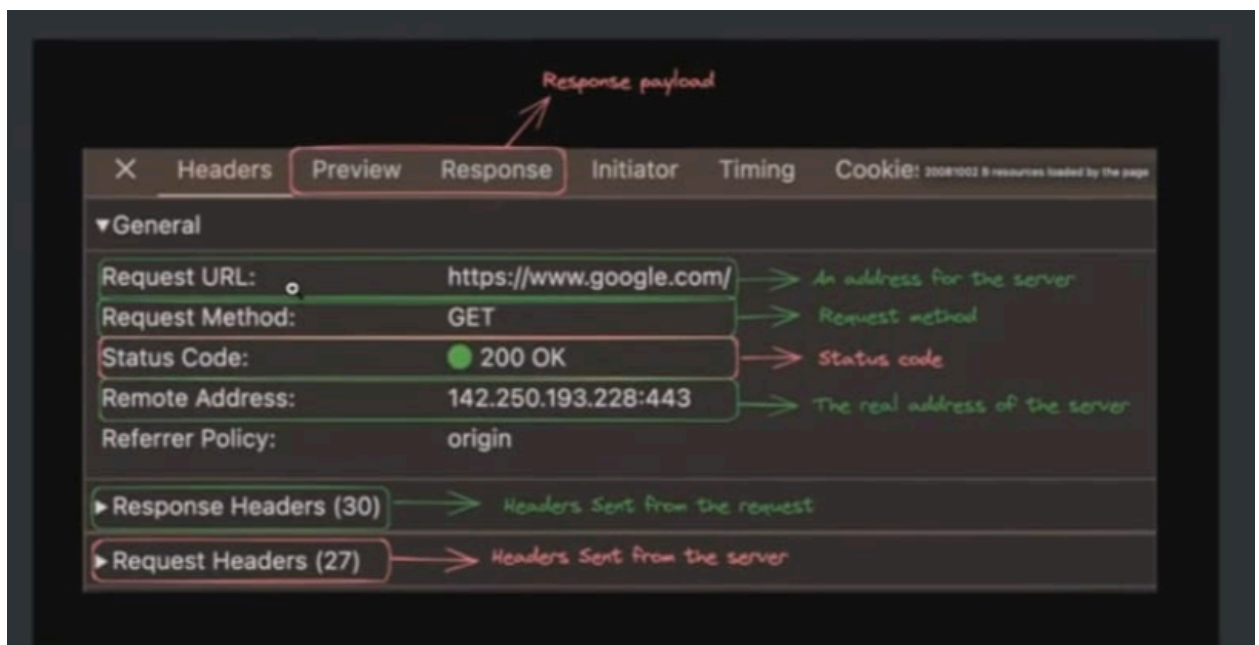
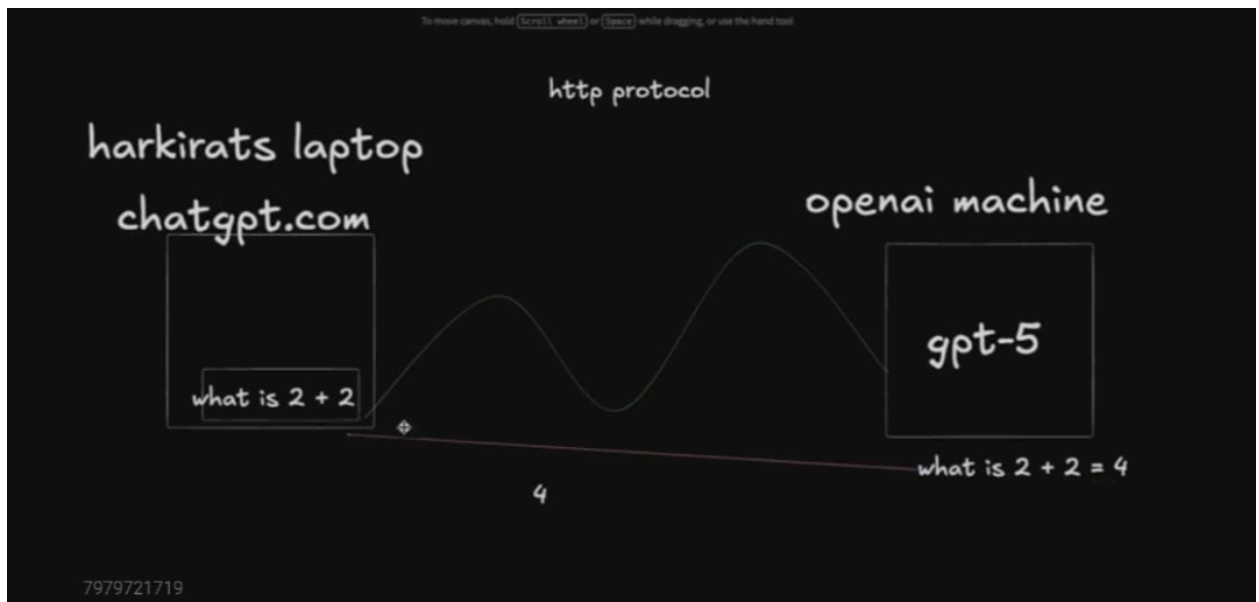
6. HTTP is a protocol used to transfer data between two machines by following a set of predefined rules.



7.



8. Here green color field is client request and red color field is server response



9. Request Response Model

90% website use http protocol only

Zoom use http, websockets protocol, webrtc

Request response model

The request-response model is a fundamental communication pattern.

It describes how data is exchanged between a **client** and a **server** or between two systems.

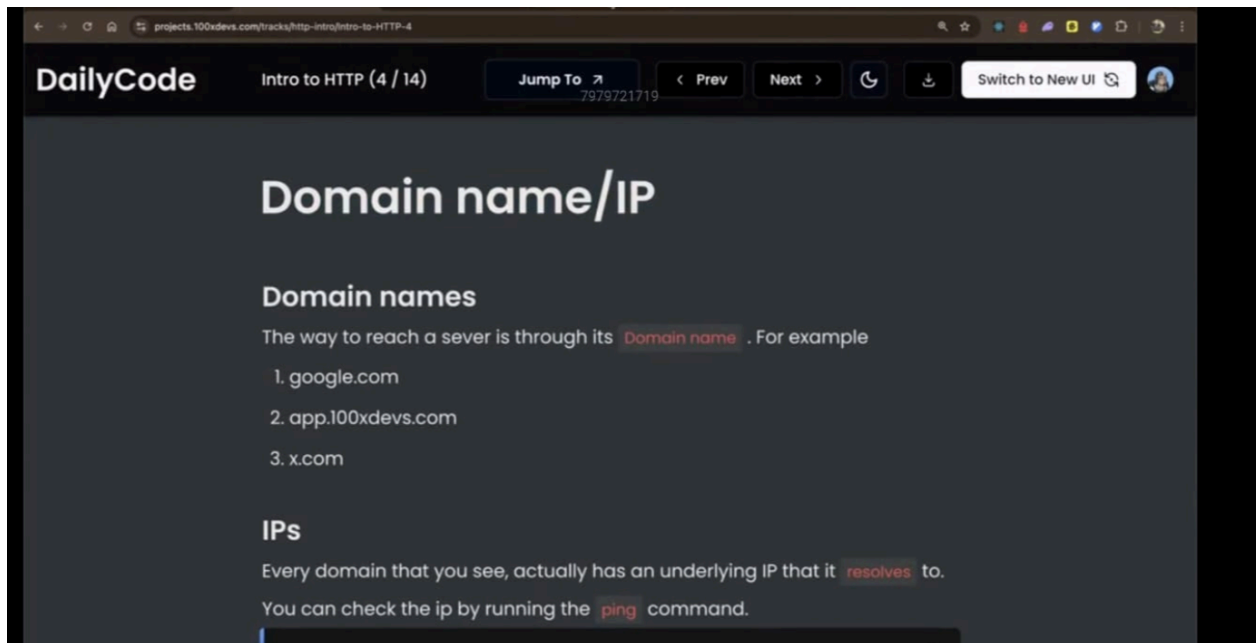


zoom -- http, websockets, webrtc



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10. Domain name/IP

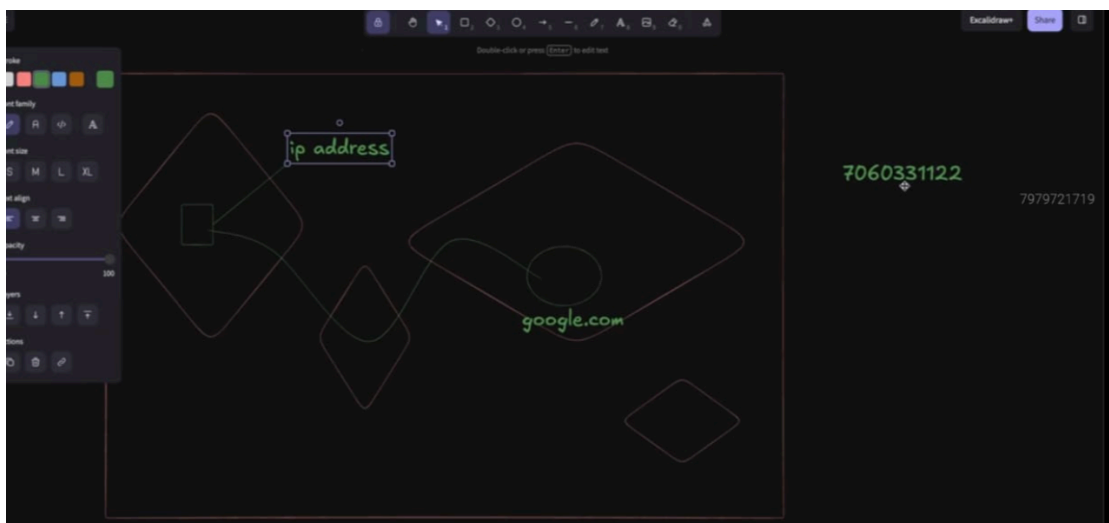
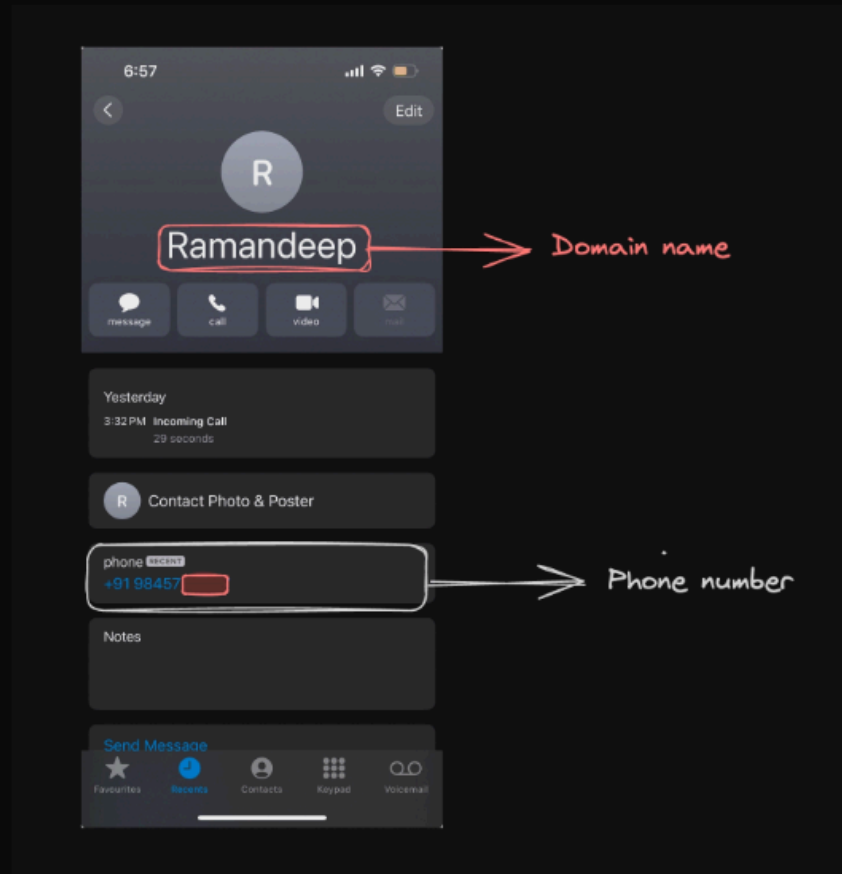


In real life, people usually remember **names**, not **phone numbers**. When we call someone, we select their name from our contacts, but the phone actually connects using their **phone number**. Similarly, on the internet, users remember **domain names** (like **amazon.com**) instead of **IP addresses**. When we send a request, the system first converts the domain name into an **IP address** using **DNS (Domain Name System)**. Then the request is sent to that IP address, and the server returns the response.

Real world analogy -

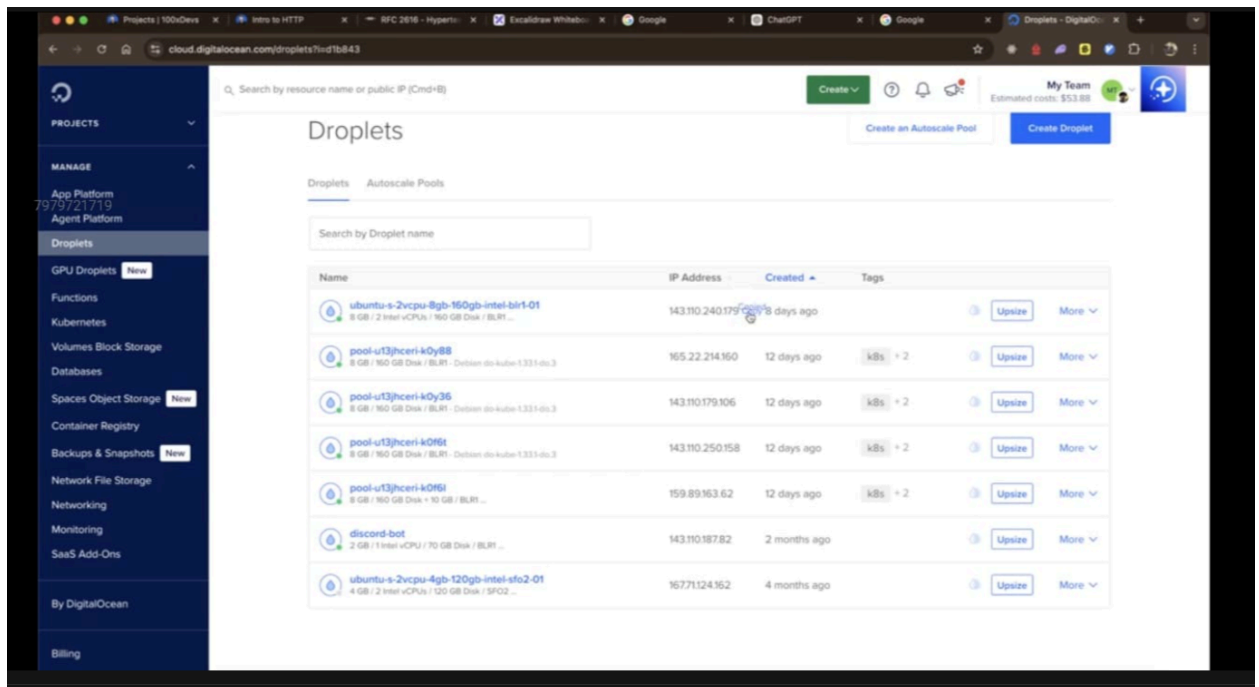
Domain name - Phone contact

IP - Their real phone number



So each machine have ip address even our laptop have ip address

Harkirat demonstrated an example using **DigitalOcean**. He showed that whenever we create a machine (called a **droplet**) in DigitalOcean, the cloud provider automatically assigns it a **public IP address**, which allows other machines on the internet to communicate with it.



Homework: Search about what is static/public ip address -> harkirat said is not easy to find these ip address only cloud can know

If you see ip address have format i.e. it follow ipv4 format


```
ipv4 -  
0.0.0.0  
255.255.255.255
```

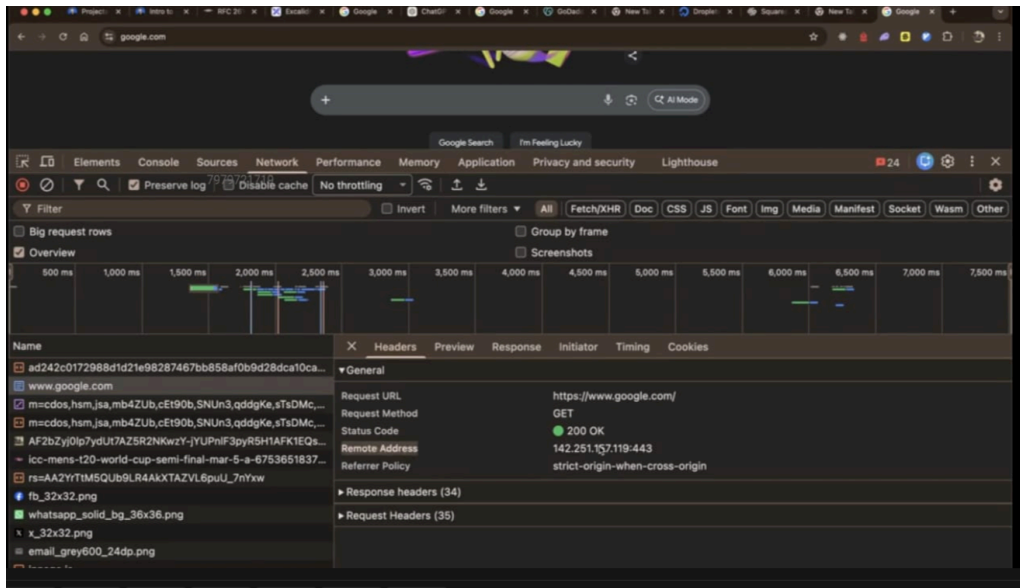
Homework: Why did ipv6 come into picture?

To get to know about ip address of any domain name:

Use: ping website name in Terminal

```
~ p  
~ ping google.com  
PING google.com (172.217.24.78): 56 data bytes  
64 bytes from 172.217.24.78: icmp_seq=0 ttl=118 time=13.497 ms  
64 bytes from 172.217.24.78: icmp_seq=1 ttl=118 time=19.810 ms
```

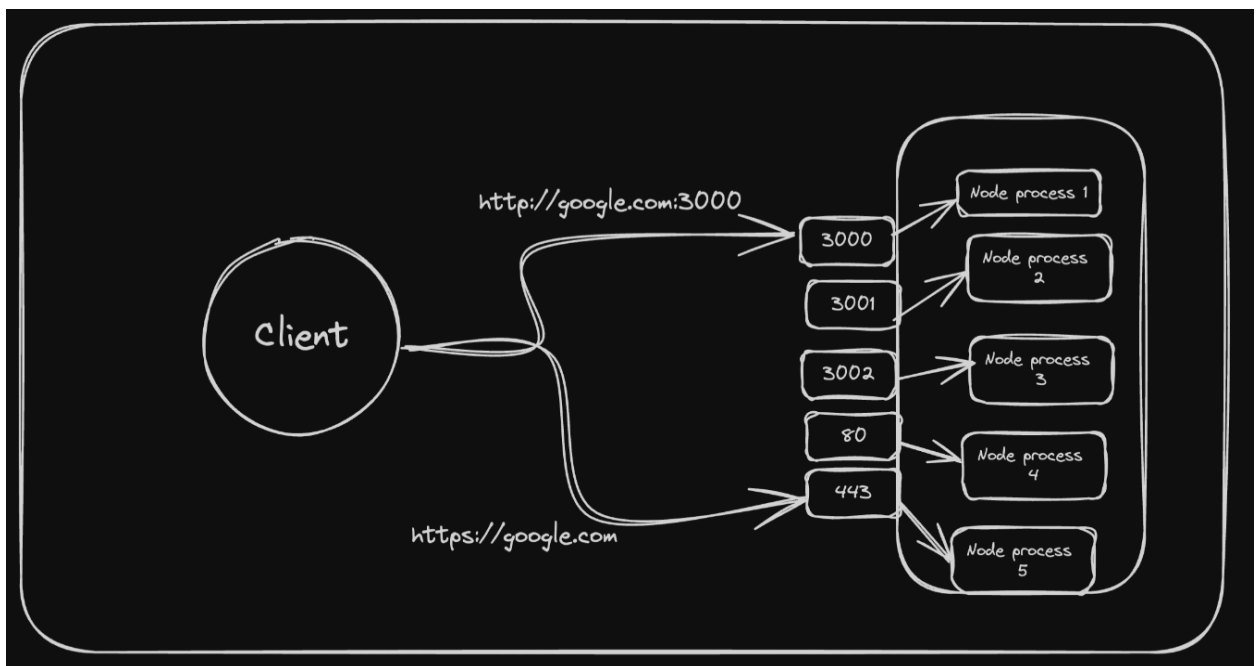
You can find IP address in network tab of each website - Remote Address

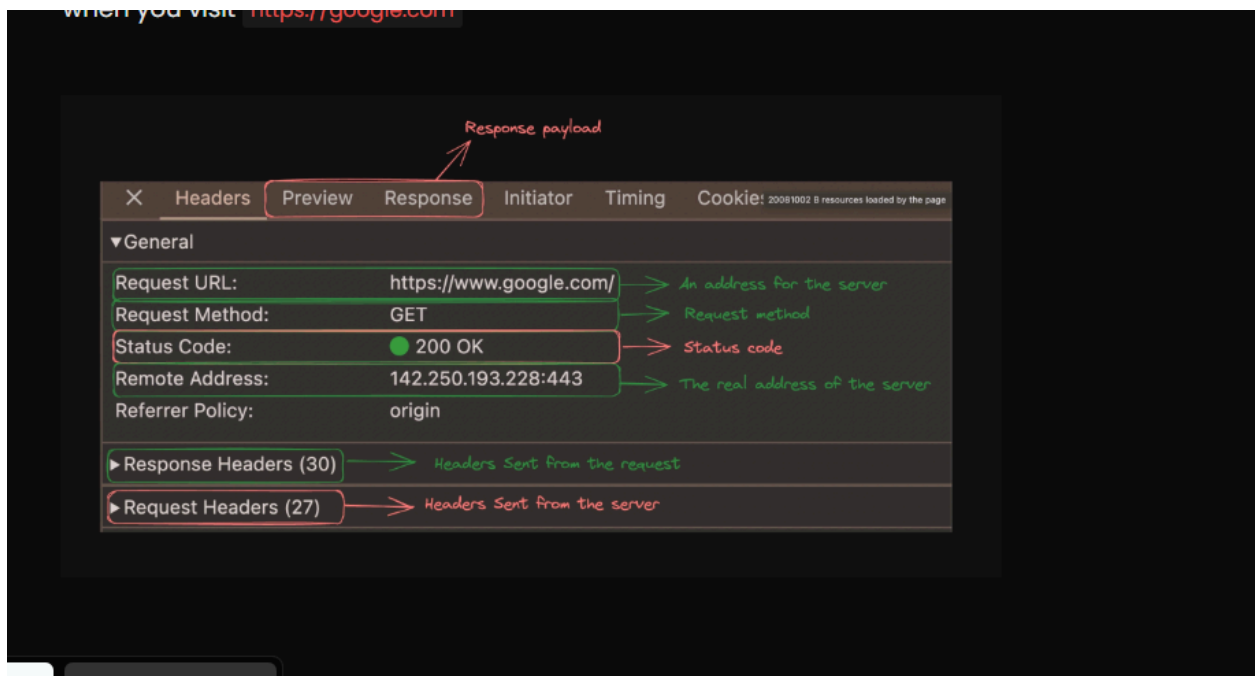


11. Ports:

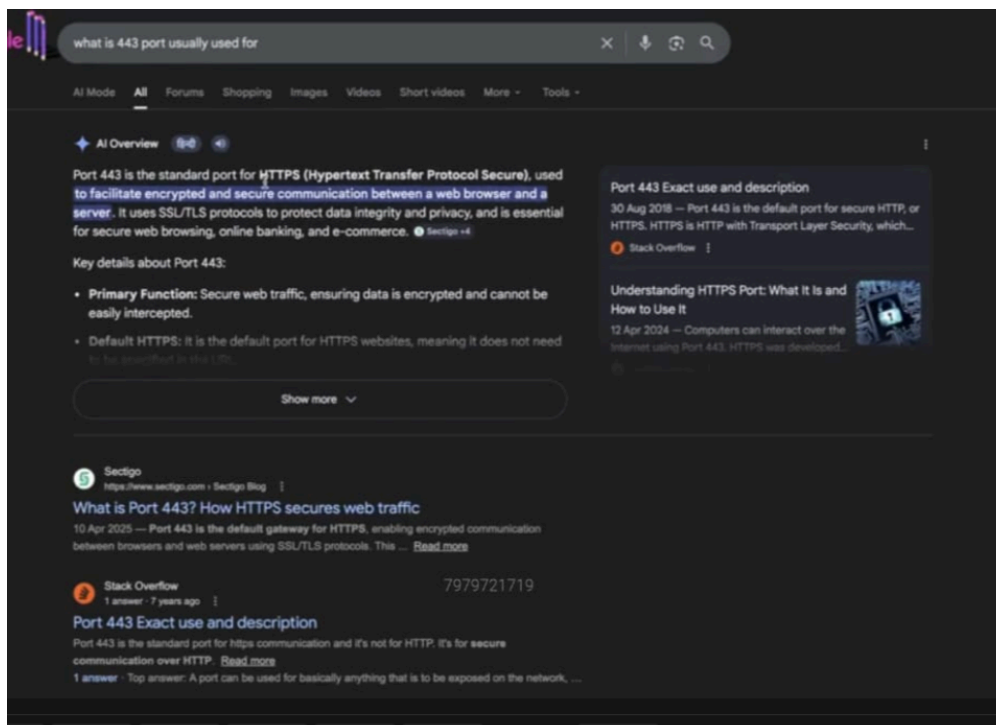
Instead of creating **five separate machines in the cloud**, we can run **multiple servers on a single machine**.

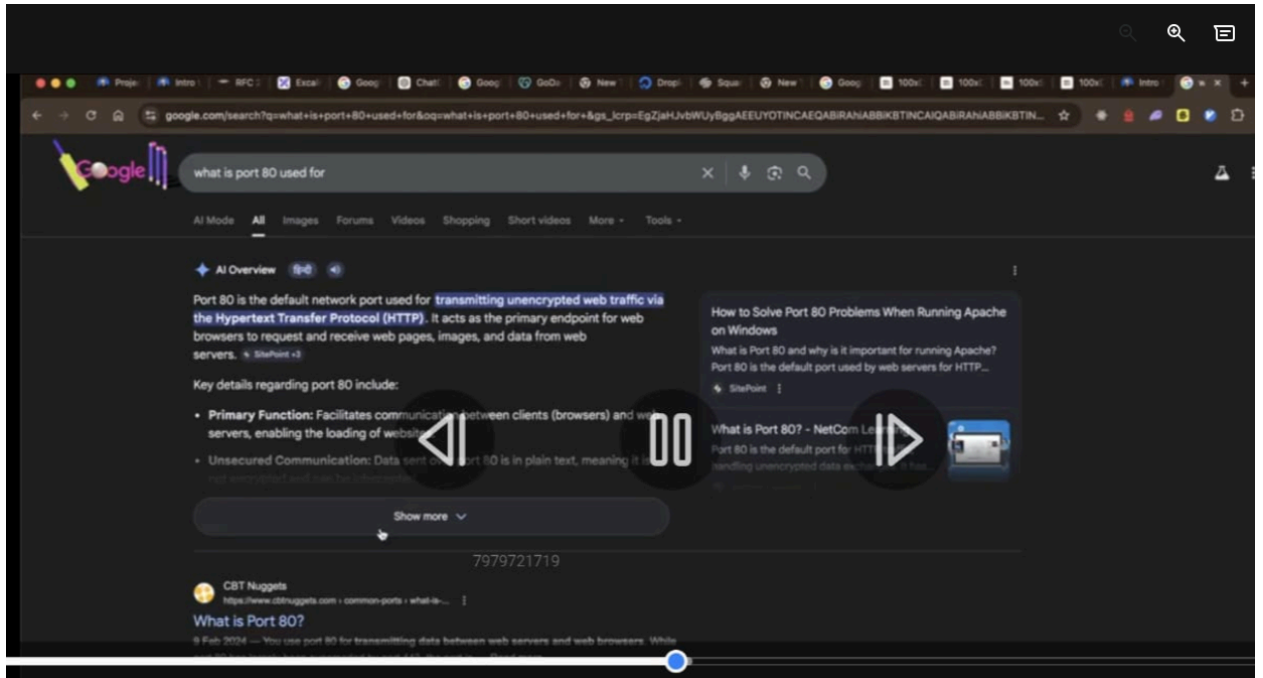
Each server listens on a **different port number**, which helps the system identify which server should handle the incoming request.



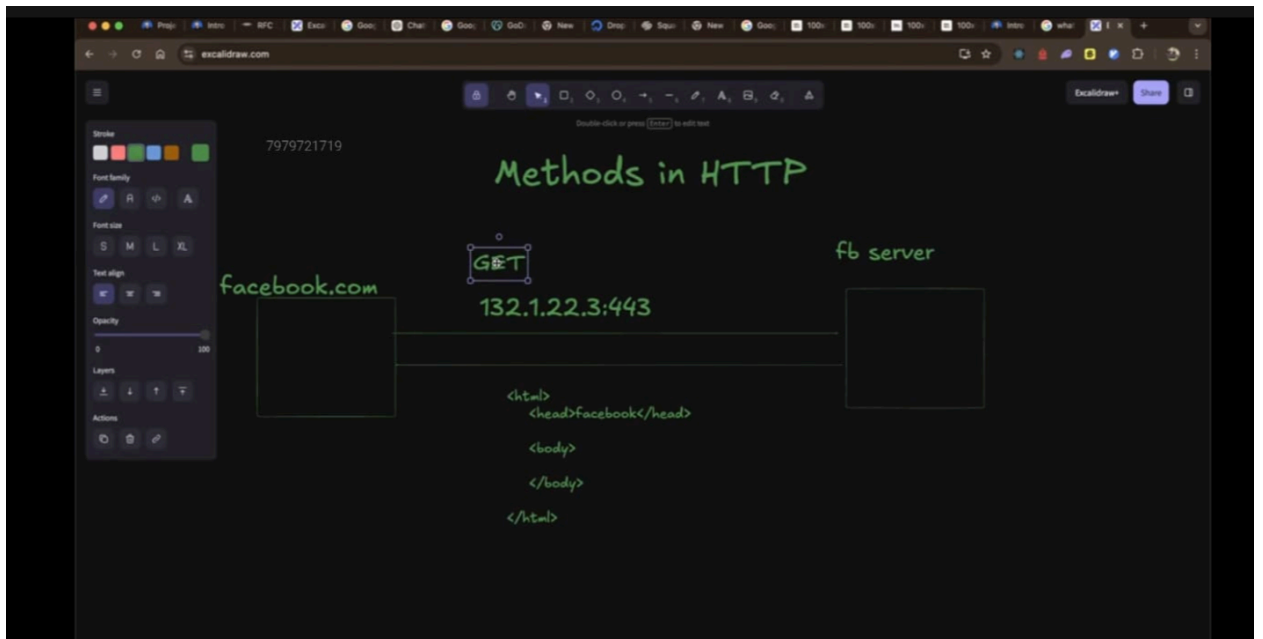


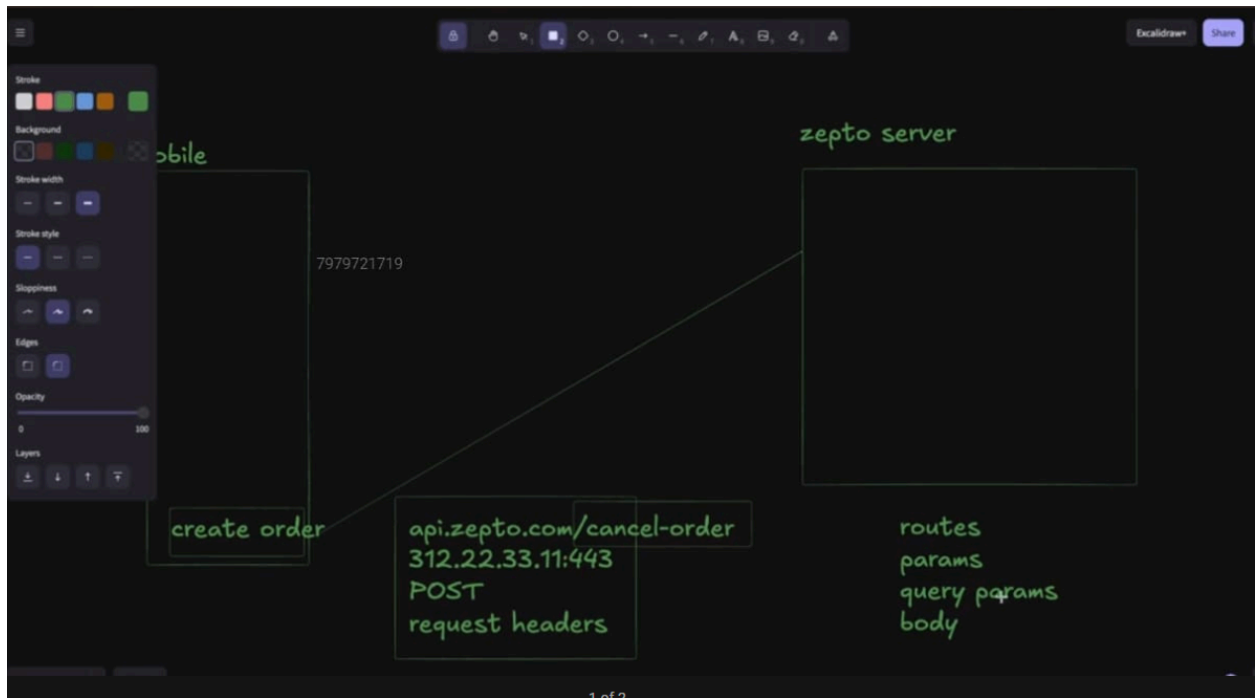
By default port of any website(https) is 443 and if you run on http is 80





12. Methods





14. Response:

<https://projects.100xdevs.com/tracks/http-intro/Intro-to-HTTP-7>

Response

The response represents what the server returns you **in response** to the request.

It could be

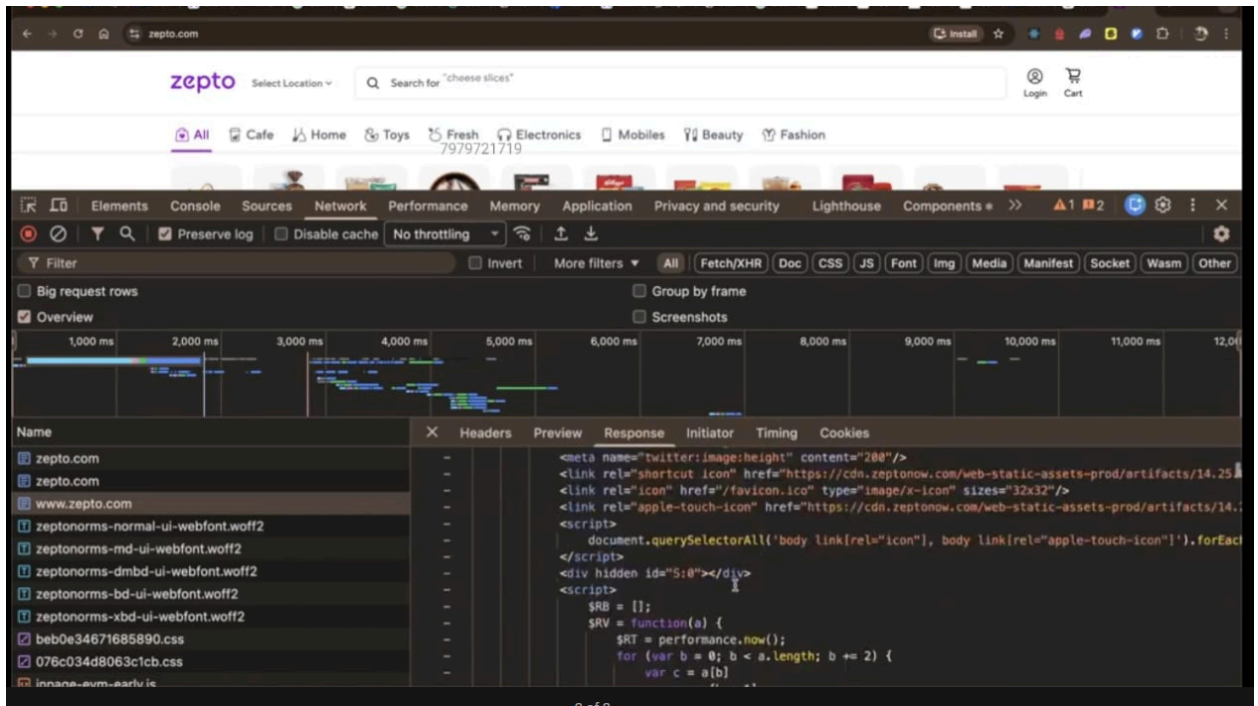
1. Plaintext data - Not used as often
2. HTML - If it is a website
3. JSON Data - If you want to fetch some data (user details, list of todos...)

JSON

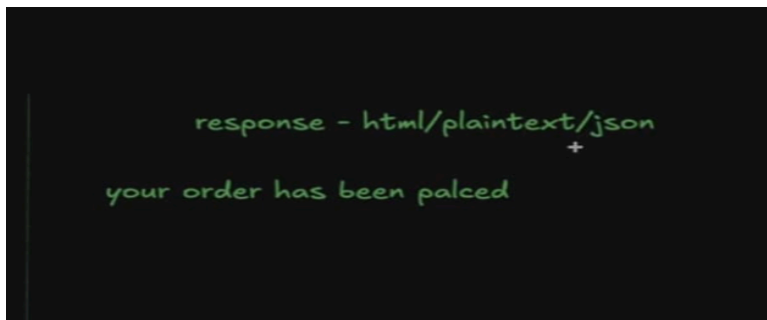
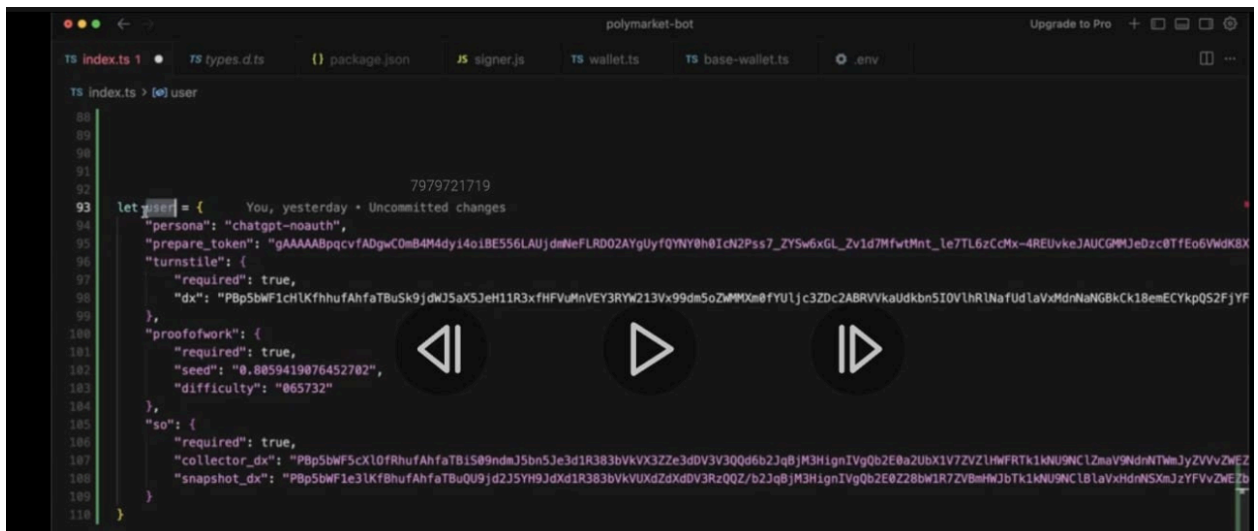
JSON stands for **JavaScript Object Notation**. It is a lightweight, text-based format used for data interchange

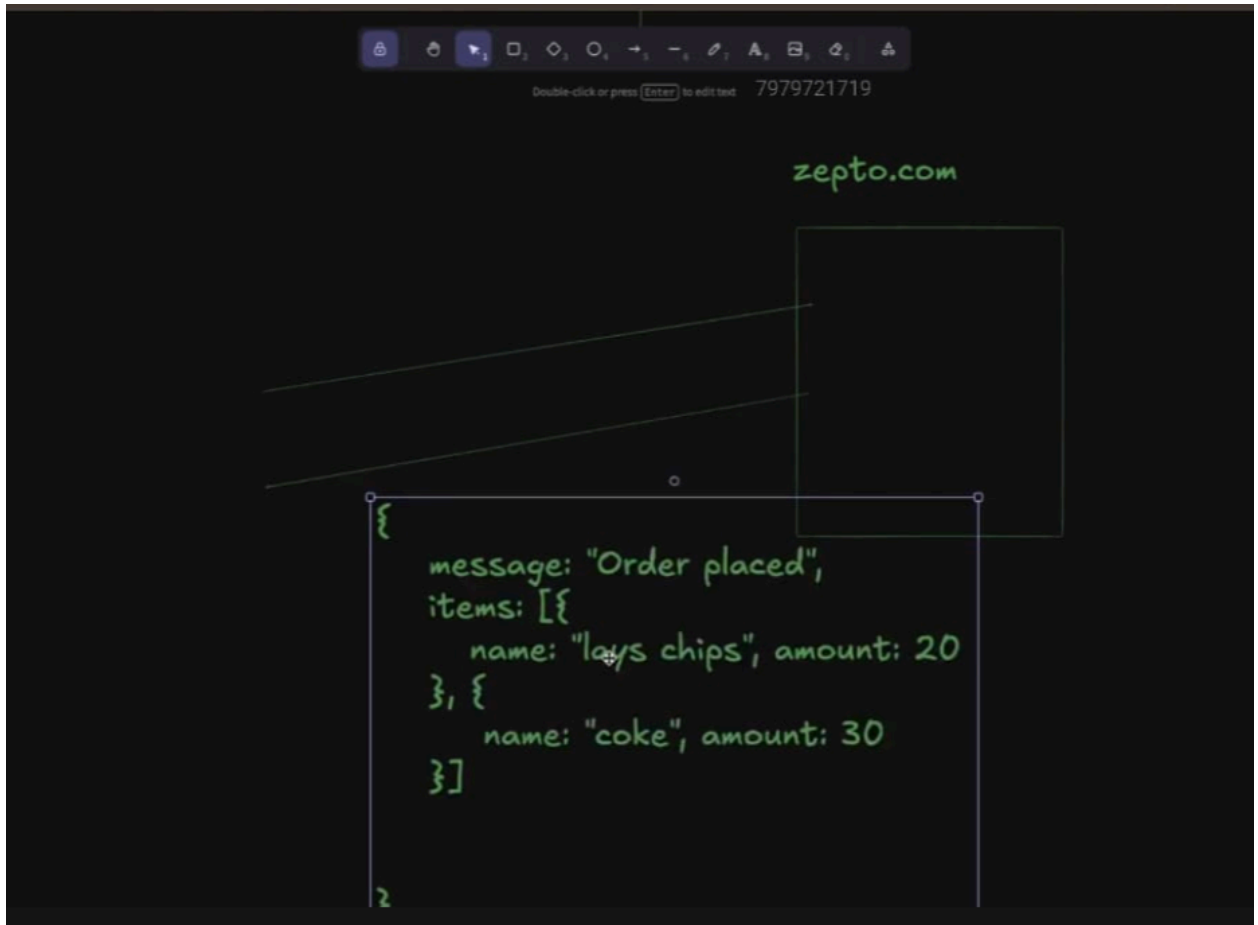
```
{
  "name": "John Doe",
  "age": 30,
  "isEmployed": true,
  "address": {
    "street": "123 Main St",
    "city": "Anytown"
  },
  "phoneNumbers": ["123-456-7890", "987-654-3210"]
}
```

X Headers Preview **Response** Initiator Timing Cookie: 2008/10/2 8 resources loaded by the page



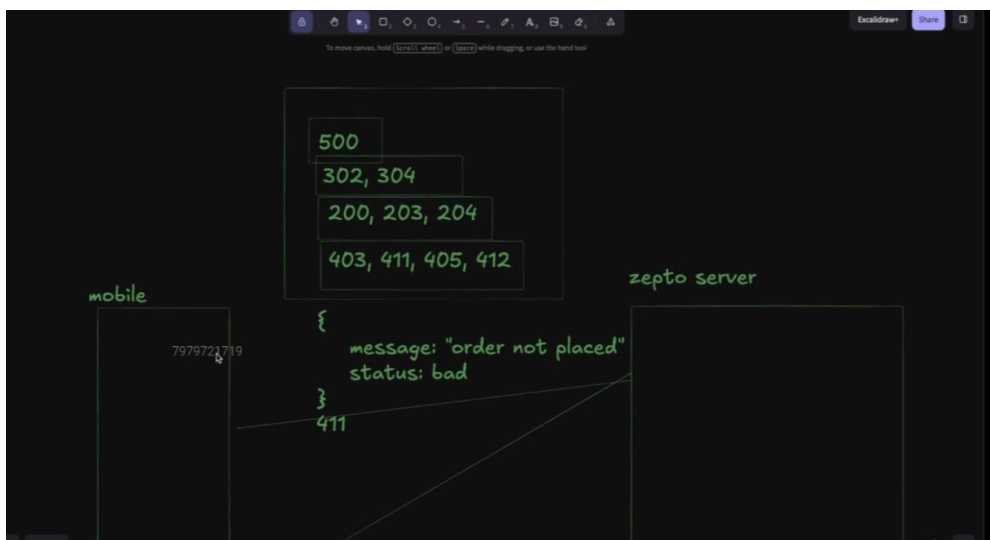
JSON Data





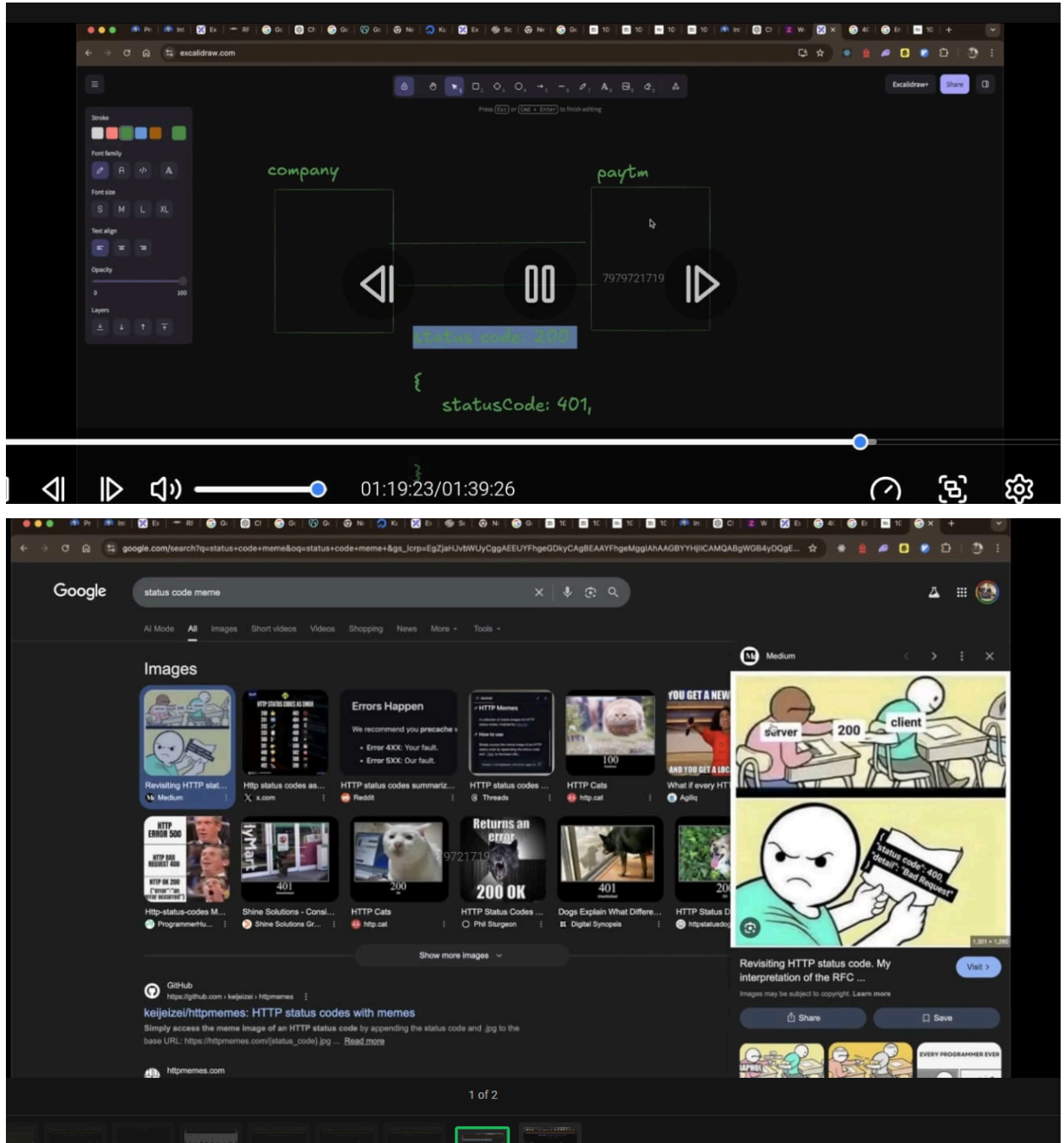
15. Status Code

<https://projects.100xdevs.com/tracks/http-intro/Intro-to-HTTP-8>



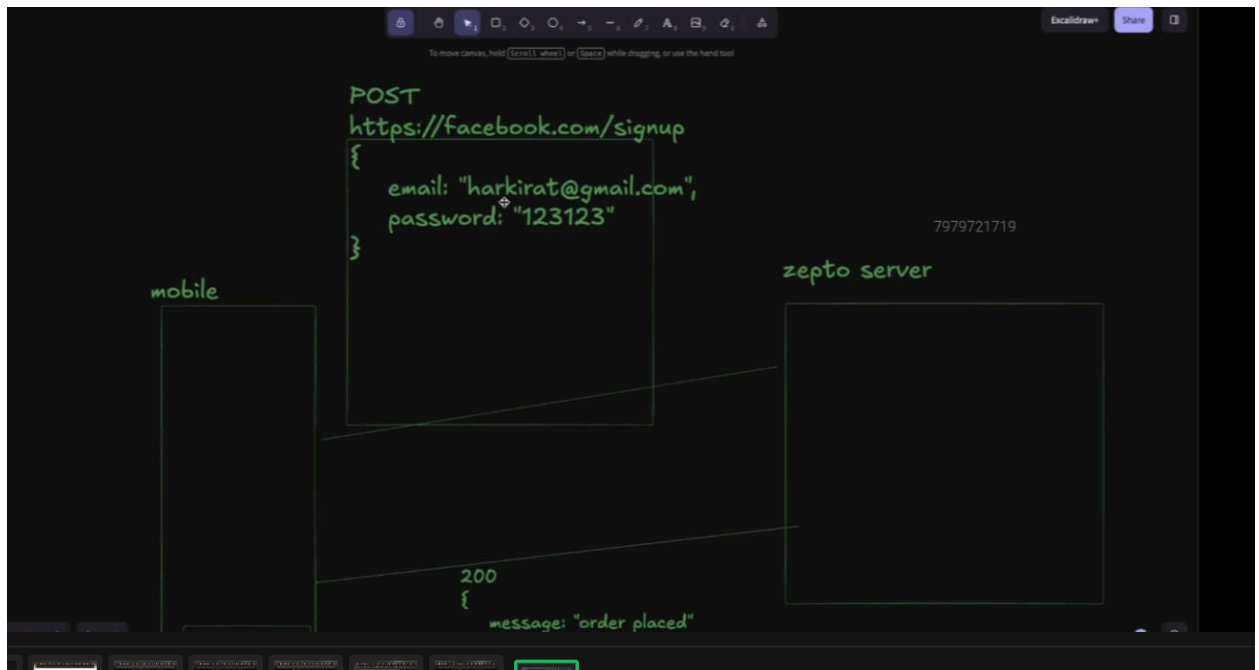
Those who write correct status code is good developer

Long before Paytm case study which return always 200 status code and in body return actual status code



16. Body

<https://projects.100xdevs.com/tracks/http-intro/Intro-to-HTTP-9>



17. Client(curl/postman/browser)

<https://projects.100xdevs.com/tracks/http-intro/Intro-to-HTTP-13>

